

ShopFast — Engineering Report

Part A: Roadmap After Test Cases | Part B: Kubernetes Migration Plan

Part A — What to Add After Test Cases

ShopFast already has unit/integration tests across most of its 16 services. Beyond tests, the highest-value additions are:

Quality & Automation

- Extend CI: Maven test/package, JaCoCo coverage, SonarCloud quality gate
- Integration tests with Testcontainers (Postgres, Kafka, Redis) for checkout→order→payment→notification
- Contract testing (Spring Cloud Contract) between gateway/consumers and producers

Observability (ELK already present)

- Micrometer + Prometheus + Grafana metrics dashboards
- Distributed tracing (Sleuth/Zipkin) across services
- Health/readiness probes and alerting

Security & Resilience

- Rate limiting & circuit breakers (Resilience4j) on the gateway
- Centralized secrets (Vault) instead of hardcoded config
- OWASP dependency scanning (Trivy/Snyk) in CI

Features & Docs

- Finish admin-service (order mgmt/status) and payment-service
- End-to-end cart→checkout→order→payment demo with frontend
- Redis caching for product/catalog reads
- Proper root README (currently 18 bytes), docker-compose/Helm, OpenAPI/Postman collection

Part B — Kubernetes Migration Plan

Current Architecture (As-Is)

Layer	Components
Microservices (16)	eureka-server, api-gateway, auth, user, product, category, inventory, cart, coupon, order, payment, review, notification, admin, elastic-service, frontend
Discovery / Identity	Eureka & middot; Keycloak 26
Messaging / Data	Kafka+Zookeeper & middot; PostgreSQL 15 & middot; Redis 7 & middot; Elasticsearch 8
Logging / Edge	ELK (Filebeat/Logstash/ES/Kibana) & middot; Nginx
Orchestration	Docker Compose

Phased Plan

Phase 0 — Prerequisites — Local cluster (minikube/Docker Desktop K8s), kubectl/helm/kustomize/skaffold, choose registry + namespace.

Phase 1 — Images — Publish common-lib once, tag by git SHA, switch to JRE base, script builds.

Phase 2 — Config — Kustomize base+overlays; ConfigMaps + Secrets; keep Eureka; inject via envFrom.

Phase 3 — Stateful Infra — Helm: PostgreSQL, Redis, Kafka+ZK, Elasticsearch+Kibana, Keycloak; reuse Filebeat/Logstash.

Phase 4 — Stateless Services — Deployments (2 replicas), readiness/liveness/startup probes, ClusterIP Services, HPA.

Phase 5 — Ingress — ingress-nginx mapping Nginx rules; frontend + api-gateway; cert-manager TLS.

Phase 6 — Observability — ELK logging; Prometheus+Grafana metrics; optional Sleuth/Zipkin tracing.

Phase 7 — Resilience/Security — Resilience4j breakers, NetworkPolicies, SealedSecrets, non-root securityContext, quotas.

Phase 8 — CI/CD — Extend GitHub workflow; Argo CD GitOps; Trivy scan.

Phase 9 — Dev Workflow — skaffold dev loop, kubectl port-forward, one-command apply -k.

Migration Order (low → high risk)

- 1. Namespace + config scaffolding
- 2. Infra (pg/redis/kafka/es/keycloak)
- 3. Eureka + api-gateway
- 4. Leaf services (product/category/user/auth)
- 5. Order/cart/payment/notification
- 6. Frontend + ingress + TLS
- 7. Observability + HPA + NetworkPolicies
- 8. CI/CD + Argo CD + prod